STROBE Statement: Checklist of Items for Observational, Functional Genomic, and Interventional Studies

Integrating CRISPRi Perturbation Responses and Transcription-Factor Binding Priors for Regulatory Network Inference in Triple-Negative Breast Cancer

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Title and Abstract	1a	Indicate study's design with a commonly used term in title/abstract	Title and Abstract explicitly indicate the study design: genome-scale single-cell CRISPRi perturbation screening, multi-omics epigenetic prior integration, causal DAG triangulation, and clinical survival validation in human TNBC cohorts.	Page 1 (Title, Abstract, Author Summary)
Title and Abstract	1b	Provide an informative and balanced summary of what was done and found	Structured abstract delineates clinical rationale (TNBC chemo-refractoriness); experimental scale (GSE224986; 95,678 cells, 11,975 sgRNAs, 819 super-enhancers); causal triangulation methodology; 6,366,350 reconstructed causal paths; prioritized master drivers (CTCF, POLR2A, EP300, JUND, MYC, SIN3A, SP1, FOS, HDAC2, MAX); enriched Hallmark pathways (EMT, E2F, G2M, Hypoxia); non-linear SHAP threshold switches; benchmarking vs GENIE3/co-expression (AUPRC = 0.875, AUROC = 0.900 vs 0.632 / 0.681); in silico driver KO validation (R^2 = 0.473, r = 0.688, p = 3.37×10^{-122}, N = 864); clinical survival in TCGA-BRCA (n = 160) and METABRIC (n = 320); and triple epigenetic combination therapy (BETi + HATi + HDACi) collapsing 93% of oncogenic circuitry.	Pages 1–2 (Abstract, Author Summary)
Introduction - Background	2	Explain scientific background and rationale for investigation	Introduction: Aggressive clinical trajectory and lack of targeted receptor therapies in TNBC; reliance on dense distal super-enhancers; mathematical proof that observational transcriptomic GRNs (GENIE3, SCENIC) suffer high false discovery rates (>45% FDR) due to single-cell dropout zero-inflation; and necessity of interventional do-calculus and functional single-cell CRISPRi.	Pages 2–4 (Introduction: Clinical Context & Epigenetic Architecture)
Introduction - Objectives	3	State specific objectives, including any prespecified hypotheses	Introduction (Objectives of This Investigation): Defines 6 primary investigative objectives: (i) construct tripartite causal GRN (TF → Enhancer → Target Gene); (ii) prioritize master drivers via PageRank and MRS; (iii) decode non-linear	Pages 4–5 (Introduction: Objectives of This

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			cooperativity via Explainable AI (TreeSHAP); (iv) benchmark against state-of-the-art GRN algorithms; (v) validate clinical prognosis in independent patient cohorts; and (vi) map synergistic epigenetic combination therapies.	Investigation)
Methods - Study Design	4	Present key elements of study design early in the paper	Materials and Methods: Comprehensive end-to-end causal machine learning workflow uniting physical chromatin occupancy (ENCODE Factorbook ChIP-seq) with randomized interventional perturbation responses (single-cell CRISPRi / dCas9-KRAB).	Pages 5–8 (Materials and Methods)
Methods - Setting	5	Describe setting, locations, relevant dates, exposure periods	Materials and Methods (Single-Cell CRISPRi Dataset Acquisition): Preclinical cellular model MDA-MB-231 (human mesenchymal basal-like TNBC); genome-scale single-cell CRISPRi Perturb-seq screen across 8 independent 10x Genomics Chromium 3' sequencing lanes (GEO accession GSE224986).	Pages 5–6 (Materials and Methods: Dataset Acquisition)
Methods - Participants / Cells	6	Give eligibility criteria, sources, and selection methods	Materials and Methods: Stringent single-cell quality control thresholds: cells retaining >= 200 detected genes and <= 15% mitochondrial reads; genes expressed in >= 3 cells; resulting in 95,678 high-quality single cells and 32,799 reliably expressed gene features.	Pages 5–6, 8–10 (Materials and Methods; Table 1; Fig 1)
Methods - Variables	7	Clearly define all outcomes, exposures, predictors, confounders	Materials and Methods: Exposure = targeted dCas9-KRAB single-guide enhancer silencing (819 somatic mutation loci); Outcome = log2 fold change and Welch t-statistic of downstream target genes; Predictors = ChIP peak confidence score C_ChIP(i, e) in [0, 1000] and causal weight W_causal; Potential Confounders = technical dropout zeros, 10x sequencing lane batches, and multi-guide epistatic crosstalk.	Pages 5–8 (Materials and Methods; Table 1)

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Methods - Data Sources	8	Give sources of data and details of measurement methods	Materials and Methods & Data Availability: GEO GSE224986 (Perturb-seq), GEO GSE223812 (enhancer coordinates), ENCODE Factorbook / UCSC encRegTfbsClustered (ChIP-seq priors across 340 TFs), TCGA-BRCA (n = 160 TNBC) and METABRIC (n = 320 TNBC) clinical cohorts, and DGIdb/ChEMBL/ClinicalTrials.gov for therapeutics.	Pages 5–8, 22–23 (Materials and Methods; Data and Code Availability)
Methods - Bias Control	9	Describe any efforts to address potential sources of bias	Materials and Methods & Discussion: Controlled technical confounders through pseudobulk aggregation, library-size depth normalization (10,000 counts per cell), randomized non-targeting controls (NTCs, n = 8,111 guides), and causal triangulation breaking backdoor confounding.	Pages 5–8, 18–19 (Materials and Methods; Discussion; S1 Text [Supplementary Note 2])
Methods - Study Size	10	Explain how study size was arrived at	Materials and Methods: 95,678 single cells sequenced across 11,975 sgRNAs (targeting 819 non-coding enhancers + NTC controls), achieving sufficient cell coverage per guide (~8 cells/guide) and per enhancer (~117 cells/enhancer) for robust statistical power.	Pages 5–6, 8–10 (Materials and Methods; Table 1; Fig 1d)
Methods - Quantitative Variables	11	Explain how quantitative variables were handled in analyses	Materials and Methods: Raw count matrices log-transformed (log1p); differential expression quantified via vectorized Welch t-test; min-max normalized physical ChIP scores; composite causal regulatory weights formulated quadratically via |log2 FC| * |t|.	Pages 6–7 (Materials and Methods; S1 Text [Supplementary Notes 1, 3])
Methods - Statistical Methods	12a	Describe all statistical methods, including confounder control	Materials and Methods: Vectorized two-sample Welch t-test (Satterthwaite df); composite causal weight formulation; personalized PageRank (teleportation probability 1-d = 0.15)	Pages 6–8 (Materials and Methods; S1 Text

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			and Master Regulatory Score (MRS) integration; Extreme Gradient Boosting (XGBoost Regressor) with TreeSHAP feature attribution.	[Supplementary Notes 3–5])
Methods - Subgroups & Interactions	12b	Describe methods examining subgroups and interactions	Materials and Methods & Results: Transcriptomic subtyping evaluated across Lehmann / Burstein clinical TNBC subtypes (MSL, BL1, BL2, LAR); clinical outcome survival analyses stratified across METABRIC (n = 320) and TCGA-BRCA (n = 160) cohorts.	Pages 5–8, 15–18 (Materials and Methods; Results; Fig 1c, Fig 8)
Methods - Missing Data Handling	12c	Explain how missing data were addressed	Materials and Methods: Handled single-cell dropout zeros using pseudobulk guide pooling and causal DAG triangulation, preventing false-positive correlation structures induced by observational zero-inflation.	Pages 5–8 (Materials and Methods; Discussion)
Methods - Sensitivity & Cross-Validation	12d	Describe sensitivity analyses and cross-validation methods	Materials and Methods & Results: Sensitivity evaluated across multiple causal weight cutoffs, comparative benchmarking against 4 alternative GRN paradigms (co-expression, GENIE3, ChIP-alone), and held-out experimental CRISPRi ground truth validation.	Pages 7–8, 15–17 (Materials and Methods; Results; Table 6; Fig 7)
Results - Participants / Cells	13	Report numbers of individuals/cells at each stage of study	Results (Single-Cell Quality Control and Global Transcriptomic Landscape): Retained 95,678 cells post-QC (median 3,412 UMIs/cell, median 1,684 genes/cell; mitochondrial fraction < 15%); 32,799 expressed genes partitioned across 8 sequencing lanes.	Pages 8–10 (Results: Single-Cell Quality Control; Table 1; Fig 1)
Results - Descriptive Data	14	Give characteristics of study participants/cells	Results: Distribution of cells across cell cycle phases (G1 = 44.2%, S = 31.8%, G2/M = 24.0%); uniform UMAP distribution across experimental replicates without batch bias; 819 targeted enhancers mapped across all human	Pages 8–10 (Results; Table 1; Fig 1a–d)

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			autosomes.	
Results - Outcome Data	15	Report numbers of outcome events or summary measures	Results: 15,059 physical TF–enhancer binding events identified across 161 sequence-specific TFs (S1 Table); 1,110,566 candidate interventional edges (|log2 FC| >= 0.25, |t| >= 2.5) filtered via vectorized Welch testing (Fig 3; S2 Data).	Pages 10–11 (Results; Table 2; Fig 2, Fig 3; S1 Table, S2 Data)
Results - Main Findings	16	Give unadjusted and adjusted effect estimates and precision	Results: Reconstructed 6,366,350 tripartite causal paths (S3 Data); prioritized master drivers: CTCF (118.7k edges, MRS = 0.986), POLR2A (110.5k, MRS = 0.983), EP300 (67.3k, MRS = 0.809), JUND (75.3k, MRS = 0.806), MYC (60.9k, MRS = 0.768), SIN3A, SP1, FOS, HDAC2, MAX (Tables 3, 4; Fig 4, Fig 5); non-linear TreeExplainer SHAP attribution ranking JUND (0.0296) and MYC (0.0174) with threshold dynamics (Table 5; Fig 6).	Pages 11–15 (Results; Tables 3–5; Fig 4–6; S3 Data, S1 Display)
Results - Benchmarking & Validation	17	Report other analyses done (sensitivity, benchmarking, validation)	Results: Triangulated Causal ML achieved AUROC = 0.8999, AUPRC = 0.8745, Top 25% Precision = 0.9013, FDR = 0.0987 vs ChIP-seq alone (AUROC 0.8890 / AUPRC 0.7228 / FDR 0.2440), Co-expression (0.7292 / 0.6814 / 0.3227), and GENIE3 (0.6698 / 0.6323 / 0.3573) (Table 6; Fig 7a,b); in silico driver KO validation yielded Pearson r = 0.688, R^2 = 0.473 (p = 3.37 * 10^-122, N = 864 genes; Fig 7c); clinical Cox survival modeling in TCGA-BRCA (n = 160) and METABRIC (n = 320) confirmed significant risk for JUND (HR = 1.88, p < 0.001), MYC (HR = 1.74, p < 0.001), FOS (HR = 1.62, p = 0.001), EP300 (HR = 1.45, p = 0.006), and MAX (HR = 1.36, p = 0.019) (Fig 7d); in silico combinatorial therapy demonstrated 93% network collapse under BETi + HATi + HDACi vs 38% under chemotherapy	Pages 15–18 (Results; Tables 6, 7; Fig 7, Fig 8)

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			(Fig 8; Table 7).	
Discussion - Key Results & Conclusion	18	Summarize key results with reference to study objectives	Discussion: Successfully triangulated 6.36M directed causal edges; resolved mathematical failure of observational GRNs (>45% FDR); established Pearlian interventional identifiability; unmasked non-linear AP-1/MYC switch dynamics; in silico knockout validation ($R^2 = 0.473$); and formulated tri-modal epigenetic combination therapy (BETi + HATi + HDACi) collapsing 93.4% of oncogenic super-enhancers while de-repressing tumor suppressors.	Pages 18–22 (Discussion)
Discussion - Limitations	19	Discuss limitations of study (potential bias, imprecision)	Discussion (Limitations and Methodological Boundaries): Reliance on preclinical in vitro cell line (MDA-MB-231) requiring future expansion into patient-derived organoids (PDOs); targeted silencing of non-coding enhancers rather than direct genetic knockouts of upstream TF coding genes; absence of single-cell Hi-C/Micro-C 3D looping data; computational rather than in vivo clinical trial testing of combination regimens.	Pages 20–21 (Discussion: Limitations and Methodological Boundaries)
Discussion - Interpretation	20	Give a cautious overall interpretation of results	Discussion & Future Perspectives: Triangulating randomized Perturb-seq interventions with physical ChIP-seq priors provides a mathematically principled, biologically validated paradigm for discovering causal oncogenic drivers and designing rational epigenetic combination therapies.	Pages 18–22 (Discussion; Future Perspectives)
Discussion - Generalisability	21	Discuss the generalisability (external validity) of study results	Discussion: Demonstrates broad applicability of the causal triangulation framework across other treatment-refractory cancers characterized by super-enhancer addiction; clinical validation across independent patient cohorts (TCGA-BRCA, METABRIC) confirms prognostic generalisability beyond	Pages 20–22 (Discussion: Future Perspectives)

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			cell line models.	
Funding & Data/Code Availability	22	Give source of funding, role of funders, and accessibility of data/code	Data and Code Availability & Declarations: Supported by National Science & Technology (NST) Fellowship (2025–2026), Ministry of Science and Technology, Government of the People's Republic of Bangladesh (Awardee: Omar Faroque). All code, notebooks, and models are available under MIT License at https://github.com/Omarrfaroqueiu79/tnbc-causal-grn; Zenodo DOI: 10.5281/zenodo.23897809; GEO accessions GSE224986 and GSE223812; Supporting Information files S1 Text through S1 Checklist.	Pages 22–23 (Data and Code Availability; Acknowledgments)

Note: This checklist adheres to the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) guidelines adapted for molecular epidemiology, high-throughput single-cell functional genomics, and interventional causal discovery. All metrics, cohort dimensions, and experimental parameters strictly match the primary manuscript text, figures, tables, Discussion, and Data and Code Availability sections.